Computer Programming

Project # 02

Write a C++ Program that computes and prints one of the Trigonometric Function using While or Do-While Loops. The program must;

a) Ask the user to select ‘S’ or ‘s’ for Sine, ‘C’ or ‘c’ for Cosine and ‘T’ or ‘t’ for Tangent. If the user enters any other character, it should print “Error message”.
b) Ask the user to enter the angle in degrees between -90 and 90. If the user enters angle outside this range, it should print “Error message”.
c) Compute the corresponding trigonometric function and displays the result.
d) Ask the user “Do you want to compute any other trigonometric function ?”. If the user enters ‘Y’ or ‘y’, it should repeat steps a – d. If the user enters another key, it should end the project.

Please note that:

1) The deadline to submit this project is Saturday, May 27, 11:59pm. Late submitted assignments even by a single minute will not be accepted.
2) You can only use While or Do-While Loops. Nested Loops are allowed.
3) You cannot use #include <cmath>.
4) You will have to use the Taylor/McLaurin series of the functions. They are given below;

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} \qquad = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \qquad \text{for all } x$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} \qquad = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots \qquad \text{for all } x$$

$$\tan x = \sum_{n=1}^{\infty} \frac{B_{2n}(-4)^n\,(1-4^n)}{(2n)!} x^{2n-1} \qquad = x + \frac{x^3}{3} + \frac{2x^5}{15} + \cdots \qquad \text{for } |x| < \frac{\pi}{2}$$

5) “n” can be up to 10.
6) Submit the solution in .cpp format. Any other form of submission will not be accepted.
7) Name the file as RegistrationNo_Project2.cpp. For example, if your registration number is 16JZELE00111, you should name the file as 16JZELE00111_Project1.cpp.
8) There is absolutely zero tolerance for cheating. If two or more students have similar projects, all the students will get Zero marks irrespective of who cheated.